1. Decision Making In Machines / Computers

***What is Data? What is Information**?

Data is a piece of raw unprocessed facts collected together for reference or analysis. Processed data is called information. Information has a logical sense.

Eg: List of students forms an example of data . When arranged either class-wise or age- wise, it forms meaningful information.

***Different types of data.**

Text data, Image data, Audio data, video data. All data types are stored as binary digits.

***What is data acquisition?**

Collecting data for a project is known as data acquisition. For any AI project to be efficient, the training data should be authentic and relevant to the problem statement. It should be unbiased to get an accurate prediction.

***Different sources of data**.

Data is the base of any AI project to be built. Data should be collected from reliable and authentic source for the accuracy of the project. The various sources for collecting data are:

Surveys- Survey is a way of collecting data from a group of people in order to gain information and insight to various topics of interest. It can be online surveys, telephonic surveys, or in person surveys.

Web Scrapping- Web scraping or Data scraping is the method of downloading information from the World Wide Web and storing it for later reference.

Sensors - Sensors can be used to collect environmental data. E.g.: data collected through sensors in autonomous vehicles.

Cameras - CCTV, web cameras, surveillance camera etc. can be used to collect visual data.

Observations- It is a method of collecting data by watching facts as they occur. E.g. : observing the growth of a plant and recording the changes.

Application Programming Interface (APIs) - APIs are a set of functions and procedures that allow one application to connect to another. APIs can be used to collect data from social media services or analysis

***How to identify relevant data for an AI project?**

We need to visualize the factors that affect the problem statement, for which we need to acquire the data.

We need to find out the parameters that will affect the problem statement.

*** What is the role of data in an AI project?**

Data plays an important role in AI project as it creates the base on which the AI project is built. For e.g.: whenever we want the AI project to be able to predict an output, it should be trained with a set of data. This data is known as **training data**. The predicted output is the **testing data.**

*** What is meant by exploring data?**

Data exploration is the process of uniformly arranging gathered data for a better understanding .Data can be organized or explored by using the techniques like:

1) data visualization –arranging data in the form of charts, tables ,graphs or diagrams.

2) Statistical Analysis- using mean ,median , standard deviation , central tendency , dispersion etc.

***What is a database?**

A database is an organized repository of an organisation's information resources ,including some raw data and procedures.

***What is deterministic problem/deterministic computing?**

A deterministic problem or algorithm is the one, whose behavior is completely determined by its inputs and the sequence of instructions given. Here the given input will always produce the same result. In deterministic computing everything is well defined and randomness or flexibility is not allowed.

Example: Finding the sum of two numbers , Rolling a dice –Each time a dice is rolled the probability of a number on six sided dice has the chance .ie:1/6 , Sorting a list of elements, digital circuit design-the behavior and output of the circuit is determined by the specific arrangement of the logic gates.

Limitations of deterministic computing are: rigidness, lack of reasoning

***What is probabilistic computing?**

Deterministic computing is a branch of computer science and artificial intelligence that emphasis on the study and application of probabilistic algorithms, models, and methods, for computation. It creates systems capable of handling uncertainty, reasoning about it and making decisions based on probabilistic world predictions.

Eg: Weather forecasting , fraud detection

***Define Automated(Automatic) System**

Automated systems perform their tasks according to the predefined rules and are deterministic in nature . ie; there is a defined problem with defined steps tp solve it.

Example: Robotic arms used to make cars in the automobile industry in the assembly line.

***Define Autonomous System**

Autonomous systems are probabilistic in nature. i.e.; they do not have predefined tasks or predefined steps to solve it. They are trained to learn from their surroundings to act independently.

Eg Self Driving Car.

***Define Decision Making.**

Decision making is the process of comparing different alternatives and coming to a conclusion according to the requirement.

Decision making by humans is subjective and is based on sentiments, feeling, belief etc which are defined by ethics ,morals and values followed by an individual.

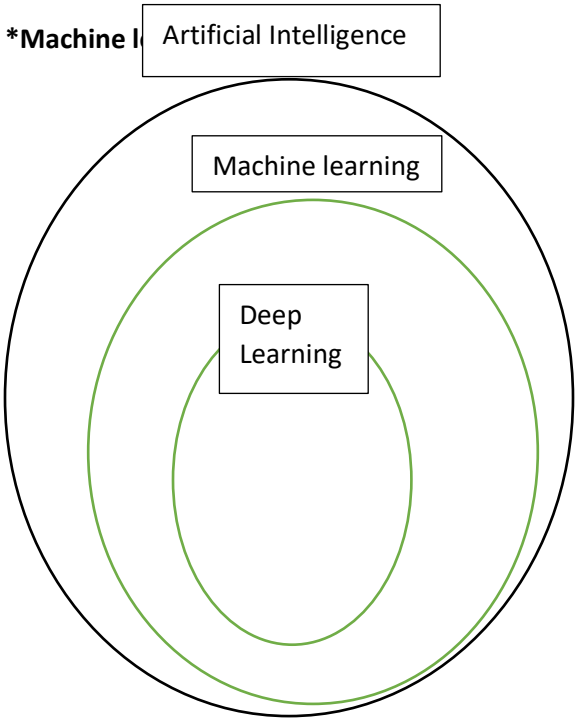
Decision making in machines is defined by learning algorithms which are objective and are programmed to be based on consequence and factual information.

***Object classification by humans and computers/machines**

Object classification by humans is made by seeing and identifying.

Object classification by machines is done with the help of informations given to it .

***Machine l**



Machine learning is an application of AI that enables a machine to learn and improve automatically from experience, gradually increasing its accuracy without the need of explicit programming. The more data a machine is given with, the more accurate the prediction will be.

Machine learning is a subset of AI. Learns things without being programmed

Deep Learning is a subset of Machine Learning. Make use of artificial neural network.

***Deep Learning**

Deep learning is a method in artificial intelligence (AI) that teaches computers to process data in a way that is inspired by the human brain. Deep learning models can recognize complex patterns in pictures, text, sounds, and other data to produce accurate insights and predictions.

***Steps in Machine Learning**

1. Data Collection: Data is critical for machine learning. The reliability and the quality of data to be assured for the model to predict accurately.

2. Data Preparation: The data collected should be prepared and made in a structured manner. This process includes data randomizing and cleaning. Cleaning involves removal of irrelevant data, duplicate data, missing values followed by restructuring of data. Once the cleaning is over the data should be split to be used as training data and testing data.

3. Model Selection: The model selection is determined based on the outcome required. It is built using various analytical techniques of machine learning which are best suited for the tasks.

4. Training the model: In this process the data prepared for training is processed by the model algorithm and understands the patterns, features and rules so as it can predict the result. Further training helps the model to make the prediction more accurate.

Testing, Evaluating and Tuning the Model: The testing data is used to check the accuracy of the Models prediction. The evaluation of the results and further tuning of the algorithm helps the model to achieve complete accuracy in the prediction.

Deployment and Prediction: Once the model is tested, it is deployed in the real world. Now the real data is fed and it should predict the outcome with great accuracy.

***Importance of Programming and Algorithms in Teaching Machines**

Programming and algorithms can be used to create models that can learn to make subjective decisions.

The instructions to the machines (models) to process the relevant data are given in the form of algorithms /programs. These are also known as Algorithmic Decision Systems (ADS).

Different methods by which programs /algorithms are used to teach the machines to make subjective decisions are:

1 Machine Learning: Machines are trained by feeding a large amount of data. From this the model learns to identify patterns and make predictions. Here the machine learns to make subjective decisions by feeding the model with data that includes both the objective facts of the situation and the subjective feeling of the people.

2 Natural Language Processing: Natural language processing (NLP) is a machine learning technology that gives computers the ability to interpret, manipulate, and comprehend human language. NLP can be used to teach machines the nuances of human language. Ie.to understand the emotional content of words and phrases.

3 Programming: Programming can be used to create algorithms that define the rules that machines will use to make decisions. This can be based on the objective facts or the subjective feelings or a combination of both.

***How do machines become intelligent?**

Machines become intelligent with the help of algorithms that are fed into them by their developers. They are trained with information that makes them intelligent to achieve the desired output. An intelligent machine have the ability to monitor its environment and then adjust its actions based on what it senses.

EVOLUTION OF COMPUTING

Binary logic which uses a system of ones and zeros (bits) to represent ad manipulate data is the basis of modern computing. The earlier computers where based on binary system and executed instructions sequentially. This system was the root of development of complex systems like AI models

Low-level Programming Language- assembly language was developed to provide more readable representation of machine code or binary code.

Operating Systems- manage resources, offer a user-friendly interface, and permit multitasking and memory management thereby making computing more accessible.

High-Level Programming Languages-Such as COBOL, C, C++ etc. were developed to facilitate the programming process and improve the readability of source code

What is a Logic Gates/Conditional Gates?

A logic gate is a device that acts as a building block for digital circuits. They perform basic logical functions that are fundamental to digital circuits. Most electronic devices we use today will have some form of logic gates in them. These gates use binary signals (0s and 1s) to perform logical functions

In a circuit, logic gates work based on a combination of digital signals coming from its inputs. Most logic gates have two inputs and one output, and they are based on Boolean algebra. At any given moment, every terminal is in one of the two binary conditions: true or false. False represents 0, and true represents 1.

Depending on the type of logic gate being used and the combination of inputs, the binary output will differ. A logic gate can be thought of like a light switch, where in one position the output is off (0), and in another, it is on (1). Logic gates are commonly used in integrated circuits(IC).

Basic logic gates

There are seven basic logic gates: AND, OR, XOR, NOT, NAND, NOR and XNOR.

The AND gate -is named so because, if 0 is false and 1 is true, the gate acts in the same way as the logical "and" operator. The following illustration and table show the circuit symbol and logic combinations for an AND gate. (In the symbol, the input terminals are on the left, and the output terminal is on the right.) The output is "true" when both inputs are "true." Otherwise, the output is "false." In other words, the output is 1 only when both inputs are 1.

Diagram and Truth Table of the AND logic gate.

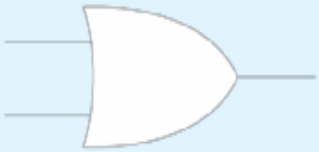


Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0
1	1	1

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The OR gate - The output is true if one or both of the inputs are true. If both inputs are false, then the output is false. In other words, for the output to be 1, at least one input must be 1.

Diagram and Truth Table of the OR logic gate.



Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	1

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The NOT gate- has only one input. A NOT gate reverses the logic state. If the input is 1, then the output is 0. If the input is 0, then the output is 1

Diagram and Truth Table of the NOT logic gate (invertor).



Input 1	Output
1	0
0	1

The NAND (Negated AND) gate - operates as an AND gate followed by a NOT gate. The output is false if both inputs are true. Otherwise, the output is true. Another way to visualize it is that a NAND gate inverts the output of an AND gate.(complement of AND gate) The NAND gate symbol is an AND gate with the circle of a NOT gate at the output.

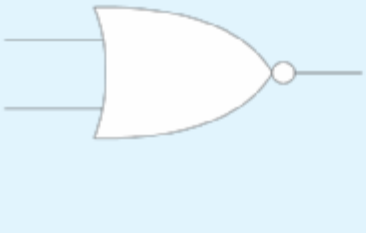
Diagram and Truth Table of the NAND logic gate



Input 1	Input 2	Output
0	0	1
0	1	1
1	0	1
1	1	0

The NOR (NOT OR) gate - is a combination OR gate followed by an inverter. Its output is true if both inputs are false. Otherwise, the output is false. (Complement of OR gate)

Diagram and Truth Table of the NOR logic gate

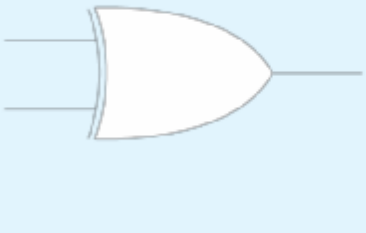


Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	0

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The XOR (exclusive-OR) gate - The output is true if either of the inputs are true. The output is false if both inputs are "false" or if both inputs are true. Similarly, the output is 1 if the inputs are different but 0 if the inputs are the same.

Diagram and Truth Table of the XOR logic gate



Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	0

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The XNOR (exclusive-NOR) gate - is a combination of an XOR gate followed by an inverter. Its output is true if the inputs are the same and false if the inputs are different.(compliment output of X-OR)

Diagram and Truth Table of the NOR logic gate



Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	1

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Complex operations can be performed using combinations of these logic gates. In theory, there is no limit to the number of gates that can be arrayed together in a single device. But in practice, there is a limit to the number of gates that can be packed into a given physical space.

Composition of logic gates- High or low binary conditions are represented by different voltage levels. The logic state of a terminal can, change as the circuit processes data. In most logic gates, the low state is approximately zero volts (0 V), while the high state is approximately five volts positive (+5 V).

Digital circuits -Logic gates can be used to make digital circuits that can do mathematical calculations, handle signals etc. This binary reasoning is the basis of computers and microprocessors.

Programming a computer- Binary reasoning is an important part of programming. Programming languages use Boolean data types (true/false) and logical operators (AND, OR, NOT)

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